

Verification phases

Phase 1 MATLAB reference



100%

50 frames, 8 algorithms, re-verified

Phase 2 HLS C-simulation



100%

4 designs, 50 frames, all bit-exact

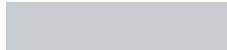
Phase 3 Synthesis + RTL co-sim



100%

4 designs, resources and exact latency

Phase 4 On silicon

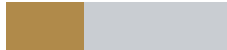


0%

needs board access

Thesis A objectives

Obj 1 Mitigate quantisation error



35%

root cause found; fix not yet implemented

Obj 2 Broaden algorithm comparison



65%

LPC and the no-prediction baseline done

Obj 3 Full compress + decompress chain



85%

both directions built and verified in simulation